SITUATIONAL TEST OF EMOTION MANAGEMENT – YOUTH VERSION

TECHNICAL REPORT, SCORING WEIGHTS AND TEST PROTOCOL

The Situational Test of Emotion Management for Youths (STEM-Y) represents a downward extension of MacCann and Roberts' (2008) Situational Test of Emotion Management (STEM). The STEM-Y was developed in order to more accurately assess emotion management skills in middle-school aged students (age approximately 11 to 15 years old). This technical report describes the item development and statistical procedures used to arrive at the final 11-item STEM-Y.

MacCann and Roberts' (2008) STEM is a research instrument designed to assess emotion management among English-speaking adults. The assessment consists of 44 multiple-choice questions which were developed based on a three-step process. First, semi-structured interviews were held with 50 adults, with qualitative analysis of these interviews used to develop scenarios. Second, a further sample of 99 individuals provided several options for managing each scenario. Qualitative analyses of these responses were used to develop response options to each scenario. Third, emotions experts (emotions researchers, life coaches, clinical psychologists, and counsellors) judged the efficacy of each response, and these expert judgments were used to score the STEM.

To develop a downward extension of the STEM appropriate for administration to children in an educational setting, two professional item writers from the Educational Testing Service (ETS) were provided with the coded interview transcripts for the development of the STEM, as well as the final list of STEM items. These item writers used these materials to develop analogous content that represented the same general themes, but was appropriate to the lives of middle school students. Sixteen items, each with four possible responses, were developed for the STEM-Y in this way. An expert scoring key was developed by asking 17 experts (7 males, 10 females) to rate the effectiveness of each response on a 6-point scale. Experts consisted of 6 emotions researchers, 7 clinical psychologists, and 4 educators who worked with middle-school-aged students.

Items were selected on the following basis. First, an index of expert agreement was calculated, representing the extent of agreement about the effectiveness of each response to the situation. Items with low levels of expert agreement were deleted from the assessment. Second, a latent class analysis of the 16 item responses was conducted, to determine whether the patterns of item responding represented classes of test-takers who were more or less effective at emotion management. That is, the expert scores on emotion management should be significantly different for different latent classes if expert scores accurately index high levels of emotion management. Third, a one-factor confirmatory factor analysis of expert-scored item responses was conducted. Items that showed low factor loadings were excluded.

Method

Procedure

This study uses data collected from 382 teen-agers and their parents. Students entering the seventh-grade were recruited from five cities in the U.S. (Atlanta; Chicago; Denver; Fort Lee, [NJ]; and Los Angeles). Each student and a caregiver were tested at a local site and compensated for participation. Students completed a proctored self-paced computerized test battery of one to two hours duration, and were prompted to take a break mid-way through the battery. During this time, parents completed a brief paper-and-pencil test battery in a separate room. All tests and protocols were approved by the ETS human ethics and fairness review committee.

Participants

A total of 382 students (51% male) from five U.S. states participated. Most of the students were 13 years of age (73.8%, M = 13.22, SD = .49, Range = 12 to 15). The majority of students lived in rural / suburban areas (62.8%). The median reported family income was between $66,000 and $80,000. The sample comprised the following self-reported ethnicities: White / Other (71.5%), Hispanic (15.2%), and African American (13.4%).

Test Battery

1. Situational Test of Emotion Management for Youths (STEM-Y), Self-Report. The STEM-Y is a downward extension of the Situational Test of Emotion Management (STEM, MacCann & Roberts, 2008). Sixteen scenarios describe emotional situations, including four alternative responses. Participants were required to select which of these four responses they would be most likely to perform in the given situation. The STEM-Y was scored according to the judgment of 17 experts. Experts were 7 clinical/counselling psychologists, 6 academics involved in emotions or emotional intelligence researchers, and 4 educators. Experts rated each of the four options for each scenario on a six-point scale, from “Very Ineffective” to “Very Effective”, with the mean expert rating used as the scoring weight (i.e., if the mean expert rating was 4.5 out of 6 for option C, a participant selecting option C would score 4.5).

2. Situational Test of Emotion Management for Youths (STEM-Y), Parent-report. The self-report version of the STEM-Y was modified so that items referred to “your child” rather than “you”. Parents were instructed to select the response that their child would be most likely to perform in the given situation. The parent-report STEM-Y was also scored according to the judgment of 17 experts.

Participants also completed several other assessments not relevant to this report.

Results

Evaluation of Expert Scoring

Expert agreement. For each of the 16 STEM-Y items, Kendall’s tau correlations between expert ratings were calculated. Two items showed low levels of agreement and were excluded from further analysis (see Table 1 for more detail). Over all ratings for the 14 remaining items, Kendall's tau between expert ratings ranged from .42 to .77 (median = .62).

Latent class analysis. Unrestricted LCA models with one to three classes were applied to *raw responses* for the remaining 14 items to detect whether classes of participants whose judgment is more or less similar to expert can be discerned. Separate models were run for self- and parent-reported raw responses, with both the BIC and the Vuong-Lo-Mendell-Rubin Likelihood Ratio test indicating a two-class solution in each model. Student and parent responses were then scored by the average ratings from 17 experts. For both self- and parent-reports, STEM-Y item means were significantly higher for class 1 (n = 286) than class 2 (n = 95) for all but one item. This item was removed from further analysis. Mean differences are reported in Table 1.

Table 1.

Item Parameters of Mean and Median Expert Agreement, and Both Self- and Parent-Report Standardized Factor Loadings and Mean Differences between Latent Classes

	Expert Agreement (Tau Correlation)		*Self-Report Item Statistics & Factor Loadings*			*Parent-Report Item Statistics & Factor Loadings*		
	Median τ	*Mean τ*	*Mean (SD)*	*Class 1 – Class 2*	*λ*	*Mean (SD)*	*Class 1 – Class 2*	*λ*
Q1	0.67	0.70	3.80 (1.05)	1.03**	**.52****	3.33 (1.00)	0.83**	**.40****
Q2	0.67	0.62	4.13 (0.66)	0.26**	.19**	3.93 (0.73)	0.39**	.23**
Q3	0.91	0.83	3.77 (0.89)	0.87**	**.40****	3.63 (0.96)	0.64**	**.30****
Q4	0.67	0.61	4.30 (0.88)	0.69**	**.40****	4.35 (0.86)	0.43**	**.34****
Q5	0.55	0.51	3.24 (0.88)	0.72**	**.35****	2.81 (1.04)	0.96**	**.45****
Q6	0.40	0.35	3.34 (0.65)	-	-	3.49 (0.59)		-
Q7	0.40	0.28	3.51 (0.59)	-	-	3.36 (0.53)		-
Q8	0.67	0.62	3.80 (0.66)	0.50**	**.35****	3.47 (0.98)	0.68**	**.36****
Q9	0.80	0.76	3.71 (1.25)	2.00**	**.64****	3.85 (1.17)	0.98**	**.54****
Q10	0.52	0.39	3.63 (0.45)	0.04	-	3.48 (0.32)	0.06	-
Q11	0.91	0.86	3.96 (0.84)	0.61**	**.44****	4.09 (0.76)	0.53**	**.47****
Q12	0.67	0.49	3.54 (0.72)	0.66**	**.38****	3.07 (0.76)	0.78**	**.49****
Q13	0.91	0.88	4.55 (0.76)	0.77**	**.43****	4.42 (0.85)	0.71**	**.45****
Q14	0.56	0.56	3.53 (0.65)	0.25**	.22**	3.45 (0.66)	0.44**	.24**
Q15	0.91	0.83	3.93 (0.75)	0.59**	**.42****	3.86 (0.93)	0.63**	**.48****
Q16	0.67	0.49	3.72 (0.58)	0.53**	**.44****	3.74 (0.55)	0.21**	.22**

Note. Salient factor loadings (>.30) are shown in bold text. Underlined items were not included in the final version of the STEM-Y.

Structural Analysis of the STEM-Y: Self- and Parent-Report

Velicer's Minimum Average Partial (MAP) test and parallel analysis were undertaken on the 14 remaining items to determine scale dimensionality of both self- and parent-reports. Both MAP and parallel analysis indicated a one-factor solution for both self- and parent-reports. One-factor CFAs were then undertaken for both self- and parent-reports. Fit was good for both self-reports (χ^2 = 87.470, *df* = 65, CFI = .951, RMSEA = .030, SRMR = .041) and parent-reports (χ^2 = 102.852, *df* = 65, CFI = .904, RMSEA = .040, SRMR = .047). All item loadings were significant, but 2 of the 13 items loaded less than .30 on the latent factor for both self- and parent-report analyses. As these items also showed low item variability, and their removal did not lower Cronbach's alpha values, these two items were removed and the CFAs re-run on the remaining 11 items. Again, fit statistics indicated good fit for both self-reports (χ^2 = 52.133, *df* = 44, CFI = .981, RMSEA = .022, SRMR = .036) and parent-reports (χ^2 = 65.387, *df* = 44, CFI = .939, RMSEA = .036, SRMR = .043). Factor loadings are reproduced in Table 1.

References

MacCann, C. & Roberts, R. D. (2008). New Paradigms for Assessing Emotional Intelligence: Theory and Data. *Emotion*, *8*, 540-551.

Scoring the STEM-Y

Table 2.

Score awarded to each response option of the 11-item STEM-Y

Question	Option	Score (out of 6)
Q1	a	2.82
	b	2.88
	c	4.71
	d	1.35
Q2	a	4.59
	b	4.29
	c	2.18
	d	3.06
Q3	a	4.76
	b	3.88
	c	1.76
	d	3.65
Q4	a	4.18
	b	1.59
	c	2.88
	d	2.47
Q5	a	4.29
	b	4.00
	c	3.12
	d	1.82
Q6	a	1.94
	b	3.94
	c	1.47
	d	4.53
Q7	a	2.18
	b	3.47
	c	4.47
	d	1.00
Q8	a	2.53
	b	4.06
	c	3.82
	d	2.24
Q9	a	4.82
	b	3.88
	c	2.00
	d	1.35
Q10	a	4.24
	b	4.29
	c	1.00
	d	2.94
Q11	a	3.76
	b	4.12
	c	3.53
	d	1.71

SITUATIONAL TEST OF EMOTION MANAGEMENT: YOUTH VERSION

Instructions

In this test, you will be presented with a few brief details about an emotional situation, and asked to choose from among four responses what you would be most likely to do in the situation.

Please note, there are no right or wrong answers. We all deal with situations in different ways. All that you need to do is answer each question honestly.

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1. You and James sometimes help each other with homework. After you help James on a difficult project, the teacher is very critical of this work. James blames you for his bad grade. You respond that James should be grateful, because you were doing him a favor. What would you do in this situation?

   (a) Tell him from now on he has to do his own homework
   (b) Apologize to him
   (c) Tell him "I am happy to help, but you are responsible for what you turn in"
   (d) Don't talk to him

2. You are nervous about a speech that you need to give to the whole school about a project you've been working on. You are worried that some of the students will not understanding your speech, as it deals with some very difficult topics. What would you do in this situation?

   (a) Work on your speech to make it easier to understand and ask for questions afterwards
   (b) Practice your speech in front of your family or close friends
   (c) Just give the speech
   (d) Be positive and confident, knowing it will go well

3. You move to a new community away from all your friends. You find that your friends make less of an effort to keep in contact with you than you thought they would. What would you do in this situation?

   (a) Make the effort to contact them, but also meet people in your new community
   (b) Get involved in community events like sports or fund-raisers
   (c) Let go of your old friends, who seem to be losers
   (d) Try to tell your old friends how you feel

4. You have been working very hard with other students on a group project. A member of your team gives you some badly written work to include in the project. There is very little time before the project is due. What would you do in this situation?

   (a) Suggest ways that the student can improve the work
   (b) Don't worry about it, just put the bad work in
   (c) Tell the teacher about the situation
   (d) Rewrite the work yourself

5. You have not spoken to your favorite cousin for months, although you were both very close when you were younger. You phone your cousin but he/she can only talk for five minutes. What would you do in this situation?

   (a) Understand that relationships change, but call your cousin from time to time
   (b) Make plans to drop by and visit your cousin in person and have a good chat
   (c) Realize that your cousin is growing up and might not want to spend so much time with the family anymore
   (d) Be upset about it, but realize there is nothing you can really do

6. You are the secretary for your school chess club. You just found out that some of the club members have been complaining because you lose things and don't show up on time. What would you do in this situation?

   (a) Point out that you are not the only one who is late
   (b) Make a list of your strengths and find a role in the club that would better suit you
   (c) Quit the club and look for another one that will appreciate you
   (d) Make an effort to be on time and keep track of things

7. After school, you usually go to the mall with a really good friend. When your friend's family moves to a different part of town, your friend stops coming to the mall. You miss these times. What would you do in this situation?

   (a) Get your friend to meet you at the mall even if it's not convenient for her
   (b) Go to the mall and hang out with other people
   (c) Suggest other activities you and your friend could do together even if they occur less often
   (d) Never talk to your friend again

8. You are very excited about starting a new club. Then a family member remarks that the club will probably only last six weeks. What would you do in this situation?

   (a) Ignore the family member's comments
   (b) Prove your family member wrong but doing everything you can to have fun and succeed at the new club
   (c) Discuss with the family member how this club might be different than past clubs you were involved in
   (d) Tell your family member he/she is wrong

9. Your best friend is moving to another state and is unlikely to come back. You have been good friends for many years. What would you do in this situation?

   (a) Make sure you both keep in contact through email, phone, or letter writing.
   (b) Spend time with other friends, and keep busy
   (c) Hope that your best friend will return soon
   (d) Forget about your best friend

10. You were always involved in soccer and are a very good player. You move to a new school in a community that has a lot of sports activities, but no soccer. You miss playing soccer, but are concerned about starting another sport. What would you do in this situation?

    (a) Talk to your parents and teachers about starting a soccer league in the new community
    (b) Think about your skills and choose another sport that would use those skills (running or kicking)
    (c) Give up on sports
    (d) Play pick-up games in an empty lot with a few friends after school

11. You are given a warning by your teacher for having entered a restricted area. You were never informed that the area was restricted and will do detention if you get two more warnings, which you think is unfair. What would you do in this situation?

    (a) Explain that you didn't know the area was restricted
    (b) Accept the warning and be careful not to go in the restricted area from now on
    (c) Take a few deep breaths and calm down about the situation
    (d) Spend a lot of time complaining to your friends about this